user1@eltex-practice2-pg1-v2:/root/Eltex-repo-2_clone/module1/task3.1$ ps -eo nlwp,comm | awk '$1 > 1 {count++} END {print "Количество процессов с несколькими потоками:", count}'

Количество процессов с несколькими потоками: 11

Top

```
Fields Management for window 1:Def, whose current sort field is %CPU
   Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
   'd' or <Space> toggles display, 's' sets sort.  Use 'q' or <Esc> to end!

* PID      = Process Id            EXE      = Executable Path
* USER     = Effective User Name   RSS      = Res Mem (smaps), KiB
* RUSER    = Real User Name        PSS      = Proportion RSS,  KiB
* PR       = Priority              PSan     = Proportion Anon, KiB
* NI       = Nice Value            PSfd     = Proportion File, KiB
  VIRT     = Virtual Image (KiB)   PSsh     = Proportion Shrd, KiB
  RES      = Resident Size (KiB)   USS      = Unique RSS, KiB
  SHR      = Shared Memory (KiB)   ioR      = I/O Bytes Read
* S        = Process Status        ioRop    = I/O Read Operations
* %CPU     = CPU Usage             ioW      = I/O Bytes Written
* %MEM     = Memory Usage (RES)    ioWop    = I/O Write Operations
* TIME+    = CPU Time, hundredths  AGID     = Autogroup Identifier
* COMMAND  = Command Name/Line     AGNI     = Autogroup Nice Value
  PPID     = Parent Process pid    STARTED  = Start Time from boot
  UID      = Effective User Id     ELAPSED  = Elapsed Running Time
  RUID     = Real User Id          %CUU     = CPU Utilization
  SUID     = Saved User Id         %CUC     = Utilization + child
  SUSER    = Saved User Name       nsCGROUP = CGRP namespace Inode
  GID      = Group Id              nsTIME   = TIME namespace Inode
  GROUP    = Group Name
  PGRP     = Process Group Id
  TTY      = Controlling Tty
  TPGID    = Tty Process Grp Id
  SID      = Session Id
  nTH      = Number of Threads
  P        = Last Used Cpu (SMP)
  TIME     = CPU Time
  SWAP     = Swapped Size (KiB)
  CODE     = Code Size (KiB)
  DATA     = Data+Stack (KiB)
  nMaj     = Major Page Faults
  nMin     = Minor Page Faults
  nDRT     = Dirty Pages Count
```

```
top - 10:57:06 up 5 days,  2:29,  3 users,  load average: 0.00, 0.00, 0.00
Tasks: 113 total,   2 running, 111 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3887.5 total,   1200.6 free,    531.3 used,   2447.8 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3356.3 avail Mem

    PID USER      RUSER     PR  NI S  %CPU  %MEM     TIME+ COMMAND
      1 root      root      20   0 S   0.0   0.3   0:13.71 systemd
      2 root      root      20   0 S   0.0   0.0   0:00.06 kthreadd
      3 root      root      20   0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root      root       0 -20 I   0.0   0.0   0:00.00 kworker/R-rcu_g
      5 root      root       0 -20 I   0.0   0.0   0:00.00 kworker/R-rcu_p
      6 root      root       0 -20 I   0.0   0.0   0:00.00 kworker/R-slub_
      7 root      root       0 -20 I   0.0   0.0   0:00.00 kworker/R-netns
     12 root      root       0 -20 I   0.0   0.0   0:00.00 kworker/R-mm_pe
     13 root      root      20   0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     14 root      root      20   0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     15 root      root      20   0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     16 root      root      20   0 S   0.0   0.0   0:00.21 ksoftirqd/0
     17 root      root      20   0 I   0.0   0.0   0:03.99 rcu_preempt
     18 root      root      rt   0 S   0.0   0.0   0:01.71 migration/0
     19 root      root     -51   0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      root      20   0 S   0.0   0.0   0:00.00 cpuhp/0
     21 root      root      20   0 S   0.0   0.0   0:00.00 cpuhp/1
     22 root      root     -51   0 S   0.0   0.0   0:00.00 idle_inject/1
     23 root      root      rt   0 S   0.0   0.0   0:02.04 migration/1
     24 root      root      20   0 S   0.0   0.0   0:00.22 ksoftirqd/1
     26 root      root       0 -20 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     27 root      root      20   0 S   0.0   0.0   0:00.00 kdevtmpfs
     28 root      root       0 -20 I   0.0   0.0   0:00.00 kworker/R-inet_
     30 root      root      20   0 S   0.0   0.0   0:00.01 kauditd
     31 root      root      20   0 S   0.0   0.0   0:00.11 khungtaskd
```

Запуск команды passwd в другом окне

```
*** System restart required ***
Last login: Tue Mar 17 10:25:59 2026 from 172.16.8.4
root@eltex-practice2-pg1-v2:~# su user1
user1@eltex-practice2-pg1-v2:/root$ passwd
Changing password for user1.
Current password:
```

Вывести все процессы для которых реальным пользователем является user1

```
top - 11:02:46 up 5 days,  2:35,  4 users,  load average: 0.00, 0.00, 0.00
Tasks: 119 total,   1 running, 118 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3907.3 total,   1206.1 free,    545.5 used,   2447.8 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3361.8 avail Mem
Which user (blank for all) user1
USER     RUSER     PR  NI S  %CPU  %MEM     TIME+ COMMAND
root     root       0 -20 I   0.0   0.0   0:00.00 kworker/R-kbloc
root     root       0 -20 I   0.0   0.0   0:00.00 kworker/R-blkcg
root     root     -51   0 S   0.0   0.0   0:00.00 irq/9-acpi
root     root       0 -20 I   0.0   0.0   0:00.00 kworker/R-tpm_d
root     root       0 -20 I   0.0   0.0   0:00.00 kworker/R-ata_s
root     root       0 -20 I   0.0   0.0   0:00.00 kworker/R-md
root     root       0 -20 I   0.0   0.0   0:00.00 kworker/R-md_bi
```

```
    PID USER      RUSER     PR  NI S  %CPU  %MEM     TIME+ COMMAND
  62015 user1     user1     20   0 S   0.0   0.1   0:00.00 bash
```

Найти процесс passwd

```
    PID USER      RUSER      PR  NI S  %CPU  %MEM     TIME+ COMMAND
  62026 root      user1      20   0 S   0.0   0.1   0:00.00 passwd
```

Выполнение команд

```
MiB Mem :   3916.0 total,   1206.0 free,    550.5 used,   2451.6 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3365.5 avail Mem
Send pid 62026 signal [15/sigterm] 15
    PID USER      RUSER      PR  NI S  %CPU  %MEM     TIME+ COMMAND
  62026 root      user1      20   0 S   0.0   0.1   0:00.00 passwd
  62027 root      root       20   0 I   0.0   0.0   0:00.01 kworker/u4:0-events_unbound
  62030 root      root       20   0 I   0.0   0.0   0:00.00 kworker/0:0
  62031 root      root       20   0 I   0.0   0.0   0:00.00 kworker/0:2-cgroup_destroy
```

```
MiB Mem :   3916.0 total,   1206.0 free,    550.5 used,   2451.6 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3365.5 avail Mem
Send pid 62026 signal [15/sigterm] 2
    PID USER      RUSER      PR  NI S  %CPU  %MEM     TIME+ COMMAND
  62026 root      user1      20   0 S   0.0   0.1   0:00.00 passwd
  62027 root      root       20   0 I   0.0   0.0   0:00.02 kworker/u4:0-events_power_efficient
  62030 root      root       20   0 I   0.0   0.0   0:00.00 kworker/0:0
  62031 root      root       20   0 I   0.0   0.0   0:00.00 kworker/0:2-cgroup_destroy
```

```
MiB Mem :   3916.0 total,   1206.0 free,    550.5 used,   2451.6 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3365.5 avail Mem
Send pid 62026 signal [15/sigterm] 3
    PID USER      RUSER      PR  NI S  %CPU  %MEM     TIME+ COMMAND
  62026 root      user1      20   0 S   0.0   0.1   0:00.00 passwd
  62027 root      root       20   0 I   0.0   0.0   0:00.02 kworker/u4:0-events_unbound
  62031 root      root       20   0 I   0.0   0.0   0:00.00 kworker/0:2-cgroup_destroy
```

```
%Cpu(s):  0.0 us,  0.5 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
%Node0 :  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3916.0 total,   1206.0 free,    550.5 used,   2451.6 buff/cache
MiB Swap:   3185.0 total,   3185.0 free,      0.0 used.   3365.5 avail Mem
Send pid 62026 signal [15/sigterm] 9
    PID USER      RUSER      PR  NI S  %CPU  %MEM     TIME+ COMMAND
  62026 root      user1      20   0 S   0.0   0.1   0:00.00 passwd
  62027 root      root       20   0 I   0.0   0.0   0:00.02 kworker/u4:0-events_unbound
  62031 root      root       20   0 I   0.0   0.0   0:00.00 kworker/0:2-cgroup_destroy
```

После отправки сигнала 9(SIGKILL), окно с passwd оставило такое сообщение

```
passwd: Authentication token manipulation er
passwd: password unchanged
user1@eltex-practice2-pg1-v2:/root$ sh
$ ^C
$ exit
user1@eltex-practice2-pg1-v2:/root$ cd
user1@eltex-practice2-pg1-v2:~$ passwd
Changing password for user1.
Current password: Killed
user1@eltex-practice2-pg1-v2:~$
```

Выполните команду vim ~/file_task3.txt и нажмите Ctrl-Z

```
Current password: Killed
user1@eltex-practice2-pg1-v2:~$ vim ~/file_task3.txt

[1]+  Stopped                 vim ~/file_task3.txt
user1@eltex-practice2-pg1-v2:~$
```

6. Выполните команду sleep 600, нажмите Ctrl-Z и выполните команду jobs

```
Current password: Killed
user1@eltex-practice2-pg1-v2:~$ vim ~/file_task3.txt

[1]+  Stopped                 vim ~/file_task3.txt
user1@eltex-practice2-pg1-v2:~$ sleep 600
^Z
[2]+  Stopped                 sleep 600
user1@eltex-practice2-pg1-v2:~$ jobs
[1]-  Stopped                 vim ~/file_task3.txt
[2]+  Stopped                 sleep 600
user1@eltex-practice2-pg1-v2:~$
```

7. Последнее задание (sleep 600) сделайте фоновым

```
[2]+  Stopped                 sleep 600
user1@eltex-practice2-pg1-v2:~$ bg 2
[2]+ sleep 600 &
user1@eltex-practice2-pg1-v2:~$ jobs
[1]+  Stopped                 vim ~/file_task3.txt
[2]-  Running                 sleep 600 &
user1@eltex-practice2-pg1-v2:~$
```

8. Измените число NICE у задания (sleep 600), сделав его равным 10

9. Проверьте, что число NICE у этого задания изменилось

```
user1@eltex-practice2-pg1-v2:~$ renice 10 -p $(pidof sleep)
62042 (process ID) old priority 0, new priority 10
user1@eltex-practice2-pg1-v2:~$ ps -o pid,nice,cmd -C sleep
    PID  NI CMD
  62042  10 sleep 600
user1@eltex-practice2-pg1-v2:~$
```

10. Сделайте задание vim ~/file_ task3.txt активным и выйдите из редактора

```
user1@eltex-practice2-pg1-v2:~$ jobs
[1]+  Stopped                 vim ~/file_task3.txt
[2]-  Running                 sleep 600 &
user1@eltex-practice2-pg1-v2:~$ fg 1
vim ~/file_task3.txt

[1]+  Stopped                 vim ~/file_task3.txt
user1@eltex-practice2-pg1-v2:~$ fg 1
vim ~/file_task3.txt
user1@eltex-practice2-pg1-v2:~$ jos
Command 'jos' not found, but there are 24 similar ones.
user1@eltex-practice2-pg1-v2:~$ jobs
[2]+  Running                 sleep 600 &
user1@eltex-practice2-pg1-v2:~$
```

11. Отправьте сигнал 15 (SIGTERM) заданию sleep 600 и выполните команду jobs

```
user1@eltex-practice2-pg1-v2:/root$ sleep 600
^Z
[1]+  Stopped                 sleep 600
user1@eltex-practice2-pg1-v2:/root$ kill -15 %1

[1]+  Stopped                 sleep 600
user1@eltex-practice2-pg1-v2:/root$ jobs
[1]+  Terminated              sleep 600
user1@eltex-practice2-pg1-v2:/root$
```

12. Создайте перехватчик сигналов SIGINT и SIGQUIT внутри командного интерпретатора, который выводит сообщение «Меня голыми руками не возьмёшь!» (используйте встроенную команду trap) и отправьте сигналы самому себе

```
user1@eltex-practice2-pg1-v2:/root$ trap 'echo "Меня голыми руками не возьмёшь!"' SIGINT SIGQUIT
user1@eltex-practice2-pg1-v2:/root$ kill -SIGINT $$
Меня голыми руками не возьмёшь!
user1@eltex-practice2-pg1-v2:/root$ kill -SIGQUIT $$
Меня голыми руками не возьмёшь!
user1@eltex-practice2-pg1-v2:/root$
```